

Towards Natural Language Reasoning and Hybrid Querying over Virtual Knowledge Graphs

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Abstract. Virtual Knowledge Graphs (VKGs) provide a semantic layer for unified access to heterogeneous, distributed data sources, but still require expertise in formal query languages such as SPARQL. In this work, we present a framework that combines VKGs with Large Language Models (LLMs) to support natural language querying and reasoning. The approach leverages LLMs for both mapping generation, linking data sources to the VKG, and natural language-to-SPARQL translation. Our preliminary findings indicate that not all user queries can be expressed as declarative queries. While factual questions can be answered through SPARQL, analytical requests, such as preference inference, require data-driven reasoning. To address this limitation, we propose a hybrid architecture that combines symbolic querying with model-based inference modules. This work highlights the opportunities and challenges of integrating declarative and statistical approaches, paving the way for more expressive natural language interfaces in complex data environments.

Keywords: Natural Language Interface · Virtual Knowledge Graphs · NL2SPARQL

1 Introduction

The widespread use of heterogeneous and distributed data sources has made unified, user-friendly data access both a challenge and an opportunity. Traditional data integration approaches, such as Local-as-View and Global-as-View [7], often rely on costly data warehousing or manual schema alignment, limiting scalability and flexibility. Knowledge graphs have emerged as a powerful paradigm for representing and querying structured data through a semantic layer, but building and maintaining centralized knowledge graphs remains resource-intensive, particularly in dynamic or decentralized environments.

Virtual Knowledge Graphs (VKGs) offer a lightweight alternative by enabling unified access to distributed data without physical integration. Through semantic mappings that connect data sources to a shared ontology, VKGs allow heterogeneous data to be queried as a single knowledge graph. However, they still require expertise in formal query languages such as SPARQL, creating a barrier for non-expert users.

Recent advances in Large Language Models (LLMs) offer new opportunities to address this limitation. LLMs can support both the automatic generation of

semantic mappings and the translation of natural language queries into formal query languages. To achieve this goal, we present a framework that integrates LLM-driven generation and natural language-to-SPARQL (NL2SPARQL) translation within a VKG setting.

However, our preliminary findings suggest that not all natural language queries can be effectively mapped to SPARQL. While factual and relational questions, such as “Which museums are open in Verona in winter?” or “Which hotels are located near the Arena?” can be directly translated into declarative queries, more analytical questions, such as “What are the preferences of tourists in Verona during winter?”, require a different form of reasoning. These queries involve aggregating across multiple behavioral observations, interpreting latent concepts (e.g., “preferences”), and potentially performing predictive or clustering analyses of user behavior. Such information is typically not explicitly represented in the underlying data sources or in the VKG itself, and therefore cannot be directly retrieved through SPARQL. Instead, it must be inferred using statistical or machine learning techniques. This observation highlights the need for hybrid approaches that combine symbolic querying (e.g., SPARQL) with data-driven inference. At the same time, this direction introduces several challenges, including determining when to trigger an appropriate analytical workflow, integrating symbolic and statistical reasoning, and designing unified interfaces that remain interpretable to end users. These findings point to a promising yet complex research direction toward more expressive and flexible natural language querying systems.

Contribution. In this paper, we propose a preliminary framework shown in Fig. 1 that leverages a VKG to provide a unified semantic abstraction across multiple, possibly heterogeneous, data sources. In particular, we exploit OnTop [4], an open-source OBDA system that exposes relational databases as virtual RDF graphs by linking ontology terms, namely classes and properties, to underlying data sources through mappings. The resulting VKG can be uniformly queried using SPARQL, which Ontop transparently translates into SQL queries executed over the relational databases. Unlike a purely NL2SPARQL pipeline, our approach introduces a reasoning step that analyzes the user input before query generation. This step determines whether the natural language request corresponds to a query over the integrated data sources or to an analytical inference task. In the former case, the input is translated into SPARQL and executed over the VKG through OnTop. In the latter case, the request is routed to an inference component, where relevant data retrieved from the underlying sources are processed by a machine learning or deep learning model according to the specific inference needs, such as preference inference. Building on this hybrid architecture, we introduce a natural language interface that supports both declarative querying and model-based reasoning over heterogeneous data. Declarative queries are handled by translating natural language requests into SPARQL, while analytical or preference-oriented requests are addressed by a dedicated model inference component. The pipeline integrates schema-aware mapping, controlled query generation, and intent-aware reasoning, ensuring that user requests are aligned

with the most suitable execution mechanism and the underlying data schema. This design bridges the semantic gap between non-expert users and formal query languages while extending the expressiveness of VKG-based access beyond factual retrieval, enabling both precise querying and inference-oriented analysis across distributed data sources.

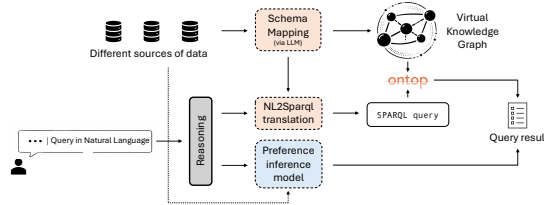


Fig. 1: Hybrid NL pipeline over a VKG integrating multiple data sources.

Running example. We consider a tourism scenario in the city of Verona, integrating heterogeneous data sources related to points of interest (PoIs): (i) a relational database containing information on cultural attractions and tourist access through the Verona Card, and (ii) an image repository containing visual content associated with PoIs. These sources differ in structure and semantics, making them a suitable case study for evaluating a VKG-based approach.

The proposed pipeline virtually integrates these sources into a unified semantic layer. Through semantic mappings, the relational data is exposed as a VKG linking each PoI to its attributes, access rules, and multimedia resources. Users can then issue natural language queries such as “Which museums can I visit with the Verona Card?” or “Show me images of monuments included in the Verona Card circuit.” The system translates these requests into SPARQL queries over the VKG, enabling user-friendly access to distributed tourism data without requiring knowledge of the underlying schemata or query language.

However, not all requests correspond to declarative retrieval tasks. For example, answering “What are the preferences of adult tourists visiting Verona during winter?” requires aggregating user behavior and inferring preferences from historical data. Such queries go beyond SPARQL-based retrieval and require integrating statistical or machine learning techniques.

Structure of the paper. The remainder of the paper is organized as follows. In Section 2, the related work is discussed, while Section 3 introduces the framework and the case study. Finally, Section 4 summarizes the findings and outlines the future directions.

2 Related Work

Ontology-Based Data Access (OBDA) and, more recently, Virtual Knowledge Graphs (VKGs) provide a well-established paradigm for integrating heterogeneous data sources through a semantic layer without requiring data materialization [14].

In this setting, OnTop is one of the most widely adopted systems for exposing relational data as a virtual RDF graph and answering SPARQL queries by rewriting them into SQL over the original sources [4].

In VKG construction, there is the problem of mappings between source schemata and the target ontology. For instance, the authors in [10] introduced a benchmark specifically designed to evaluate automatic mapping generation in relational-to-ontology integration scenarios. Recent work has systematized recurrent solutions into a catalog of conceptually grounded mapping patterns for VKGs, highlighting that mapping design, validation, and maintenance remain key practical challenges [5]. Likewise, LLM4VKG [15] has been proposed to leverage large language models for automating VKG construction.

A complementary line of research investigates natural-language access to knowledge graphs and the translation of user questions into executable graph queries. Some preliminary studies rely on template-based approaches that map natural-language questions to SPARQL query structures [13]. With the widespread use of large language models, this problem has been revisited, showing that LLMs can generate competitive SPARQL queries from natural language, even with in-context learning and without task-specific fine-tuning [6]. At the same time, recent work has shown that LLM-based semantic parsing is still affected by faithfulness and hallucination issues, especially when generating entities, relations, or URIs that must match the target graph exactly [16,11]. The task of identifying the correct path for managing a user’s natural language query aligns with recent work on intent detection and query routing, where LLMs are used to infer user intent and dispatch queries to the most suitable processing pipeline. For example, in [2], the authors demonstrate how LLMs can effectively classify intents, enabling the distinction between declarative queries and analytical tasks such as recommendation or preference inference.

Compared with these works, our framework combines all the above-mentioned aspects: LLM-assisted intent detection, the construction of a VKG across multiple heterogeneous data sources, and the translation of natural-language questions into executable SPARQL queries over that virtual semantic layer.

3 Framework

In this section, we present the proposed framework and describe its components in detail. As illustrated in Fig. 1, we consider a setting in which data is distributed across heterogeneous sources, potentially adopting different storage technologies and schemas, while remaining in their original sources and without requiring data replication or centralization. The goal is to enable users to query such data through natural language, without requiring knowledge of the underlying data structures, query languages, or inference models.

The Virtual Knowledge Graph and the mappings with data sources. The first step aims to define a unified, abstract representation of the data. In our approach, we adopt a Virtual Knowledge Graph (VKG) that provides a semantic abstraction layer over heterogeneous data sources without requiring materialization. The

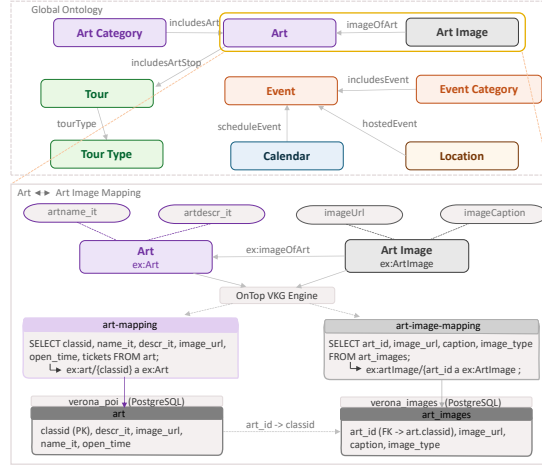


Fig. 2: OBDA mapping architecture for the Verona PoIs Virtual Knowledge Graph, illustrated for the Art-ArtImage sub-schema.

mapping between the original data sources and the VKG is automatically generated using a prompt-based approach built on GPT-4.1 mini [9]. The model is leveraged to infer semantic correspondences between source schemas and the VKG ontology. These mappings are formulated following the Ontology-Based Data Access (OBDA) paradigm [4]. Each mapping rule specifies (i) a source query, expressed in SQL, that retrieves a set of tuples from the underlying relational database, and (ii) a target template, expressed in the RDF mapping language, that instantiates the ontology terms (e.g., classes, object properties, and datatype properties) from the retrieved tuples.

Fig. 2 provides a two-level view of the resulting Virtual Knowledge Graph. The upper layer shows the global ontology structure, comprising the classes **Art**, **ArtImage**, **ArtCategory**, **Tour**, **TourType**, **Event**, **EventCategory**, **Calendar**, and **Location**, together with the object properties that relate them. The lower layer focuses on the **Art-ArtImage** sub-graph, showing the two OBDA mapping rules, the corresponding SQL source queries, and the relational tables from which the virtual triples are generated. This diagram illustrates how OnTop exposes data from two physically separate PostgreSQL databases as a coherent, queryable RDF graph without requiring data replication.

Multi-source integration. A key feature of the present setting is that the data sources span multiple, physically separate databases. In our Verona tourism case study, the main relational database (`verona_poi`) stores descriptive and operational data about PoIs, while a second database (`verona_images`) stores images associated with those PoIs. Cross-database references are resolved at the mapping level: the `art-image-mapping` rule joins the `art_images` table on the foreign key `art_id`, which corresponds to `art.classid` in the primary database, and instantiates the `ex:ArtImage` class with an `ex:imageOfArt` object

property linking each image to its associated `ex:Art` instance. At runtime, OnTop translates the incoming SPARQL query into federated SQL that spans both databases, exploiting PostgreSQL Foreign Data Wrappers (FDW) to materialize the cross-database join transparently.

Natural Language query reasoning. Our framework analyzes the user’s natural language request to determine whether it corresponds to a factual query over the integrated data sources or to an analytical inference task. In the first case, the expected output can be obtained by querying the data sources through the VKG, without requiring additional processing beyond query translation and execution. These requests are therefore routed to an NL2SPARQL component, which generates a SPARQL query to be executed over the VKG. In the second case, the request cannot be satisfied by simply retrieving explicitly represented data. Instead, it requires a more complex analytical process to derive implicit information from the available data. These requests are routed to a model-based inference component, where the relevant data are processed using a machine learning or deep learning model according to the specific task.

From text to SPARQL. To translate natural language queries into SPARQL, we integrated SPARQL-LLM [12], a competitor in the Text2SPARQL challenge, into our pipeline. It is an open-source, triplestore-agnostic framework for translating natural language queries into SPARQL. The approach leverages standardized metadata automatically extracted from SPARQL endpoints. Two main categories of metadata are required:

1. pairs of SPARQL query examples and their natural language descriptions, encoded as RDF graphs and annotated with SHACL-based vocabularies.
2. a data-aware schema that captures the structural characteristics of the underlying dataset, potentially enriched with frequency indicators. To obtain this information, SPARQL-LLM retrieves VoID [1] descriptions from each endpoint.

The overall data flow of SPARQL-LLM consists of a multi-stage pipeline. The process begins with question decomposition, in which the input natural language query is analyzed to extract potential sub-questions and identify relevant schema classes. Subsequently, embeddings are computed for both the extracted sub-questions and the identified classes. These embeddings are then used to perform similarity-based retrieval of relevant contextual information, including related query examples and schema elements.

Once the SPARQL query is generated, it is executed over the Virtual Knowledge Graph through OnTop [4]. OnTop is an Ontology-Based Data Access (OBDA) system that enables querying relational data as if it were exposed as an RDF graph, without requiring data materialization. It achieves this by rewriting SPARQL queries into equivalent SQL queries based on predefined mappings between the ontology and the underlying data sources.

In our framework, the generated SPARQL query is then provided as input to OnTop, which handles the translation and execution over the original data sources. The results are then returned to the user in a structured format consistent with the semantics of the VKG.

Model-based inference. Analytical and preference-oriented questions often require inferring latent information that is not explicitly encoded in the underlying data sources or the semantic layer.

In these cases, the framework orchestrates a dedicated inference workflow that combines data retrieval with machine learning or deep learning models. The retrieved data are processed to infer behavioral patterns, preferences, or predictive insights, and the resulting output is returned to the user as the answer to the original query.

As an initial step, we consider a lightweight preference inference module to derive aggregate tourist preferences from interaction and contextual data, currently limited to temporal data. Given a natural language request such as “What are the preferences of tourists visiting Verona during winter?”, the framework first identifies the relevant temporal context, such as the season. It then retrieves corresponding data from the underlying sources, including visits through Verona cards. In addition to these data, a simple analytical model, such as frequency-based ranking, is applied to identify recurrent patterns in tourist behavior. For example, the module may infer that winter visitors tend to prefer museums included in the Verona Card circuit or monuments located near the city center [3]. The output is therefore not a direct fact retrieved from the VKG, but an inferred preference profile derived from observed behavioral patterns.

3.1 Case Study

We evaluate the proposed framework on a real-world case study in the tourism domain, focusing on Points of Interest (PoIs) in Verona, Italy.

The dataset, introduced in [8], includes heterogeneous information for each PoI, including descriptive metadata, events, tourist affluence data spanning 2014 to 2022, and daily weather conditions. Furthermore, geospatial information, such as the location of PoIs and tourist routes, is stored, enabling the representation and querying of spatial data. In addition to this information, a separate CSV file collects links to images associated with each PoI. This setting provides a representative example of heterogeneous data sources with different schemas and modalities, making it particularly suitable for evaluating the effectiveness of our approach for integrating and querying data via natural language.

To better illustrate the behavior of the proposed system, we developed a web application that allows users to observe the execution of natural language queries on the VKG. More specifically, Fig. 3a shows the query formulation phase, in which a user-provided natural-language question is processed through the entire translation and execution pipeline. Fig. 3b presents the corresponding query results retrieved from the heterogeneous data sources integrated within the VKG.

Preliminary Evaluation. To assess the feasibility of the proposed framework, we conducted a preliminary qualitative evaluation using a set of 50 natural language questions over the Verona tourism dataset. The question set comprises 40 factual questions expected to be answered via SPARQL over the VKG, and 10 analytical questions requiring model-based inference. The evaluation focuses on two aspects:

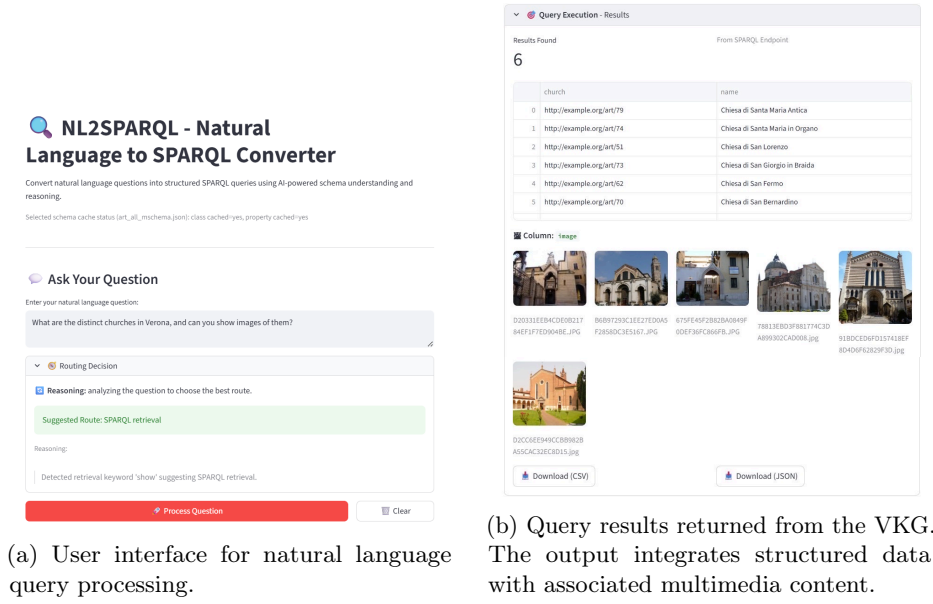


Fig. 3: Overview of the NL query interface and the generated VKG results.

(i) the correctness of the routing decision, i.e., whether the framework correctly identifies whether a question should be directed to the NL2SPARQL component or to the inference module; and (ii) the quality of the generated SPARQL queries for the factual subset. Table 1 summarizes the evaluation results across the two execution paths.

Path	Component	Questions	Correct	Accuracy
Factual	Routing (NL2SPARQL)	40	40	100%
	SPARQL correctness	40	28	70%
Analytical	Routing (Inference)	10	10	100%
Total	Routing (All Questions)	50	50	100%

Table 1: Preliminary qualitative evaluation of the framework.

Routing accuracy. The reasoning component, implemented as a prompt-based classifier using GPT-4.1 mini, correctly routed all 50 questions to the appropriate execution path. All 40 factual questions were correctly assigned to the NL2SPARQL component, and all 10 analytical questions were correctly directed to the model-based inference module. While the dataset is small and the evaluation is qualitative, this result indicates that the prompt-based intent classifier is effective at distinguishing between declarative and analytical user requests in the tourism domain.

SPARQL generation quality. For the 40 factual questions routed to the NL2SPARQL component, we evaluated the correctness of the generated SPARQL queries by executing them over the VKG using SPARQL-LLM. Of the 40 queries, 28 (70%) returned correct results. The remaining 12 queries produced either incorrect results, empty result sets due to entity-linking failures, or syntactically invalid SPARQL queries.

This simple evaluation highlights how the system supports non-expert users by exposing both the intermediate reasoning steps and the final integrated results.

4 Conclusion and Future Work

This work presents a preliminary framework that combines VKGs with LLMs to enable natural language querying across heterogeneous and distributed data sources. The proposed approach includes an intent-aware reasoning step that distinguishes between factual queries, which can be answered through SPARQL over the VKG, and analytical requests, which require model-based inference.

The Verona tourism case study highlights the potential of the proposed hybrid architecture. By combining a VKG with a natural language interface, the system enables non-expert users to query heterogeneous data sources. Preliminary results show effective handling of both factual and analytical queries, as well as promising performance on NL2SPARQL.

These findings highlight the potential of combining symbolic querying with data-driven inference to move beyond factual retrieval in VKGs. Future work will investigate intent detection and support for advanced analytical tasks, such as recommendations.

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Declaration of use of Generative AI

During the preparation of this work, the authors used Grammarly to improve the grammar and readability of the manuscript. The authors take full responsibility for the final content of the manuscript and the associated software artifacts.

References

1. Describing linked datasets with the void vocabulary. <https://www.w3.org/TR/void/>.
2. G. Arora, S. Jain, and S. Merugu. Intent detection in the age of llms. In *Proceedings of the 2024 conference on empirical methods in natural language processing: Industry track*, pages 1559–1570, 2024.

3. A. Belussi, A. Cinelli, A. Dalla Vecchia, S. Migliorini, M. Quaresmini, and E. Quintarelli. Forecasting poi occupation with contextual machine learning. In *European Conference on Advances in Databases and Information Systems*, pages 361–376. Springer, 2022.
4. D. Calvanese, B. Cogrel, S. Komla-Ebri, R. Kontchakov, D. Lanti, M. Rezk, M. Rodriguez-Muro, and G. Xiao. Ontop: Answering sparql queries over relational databases. *Semantic Web*, 8(3):471–487, 2016.
5. D. Calvanese, A. Gal, D. Lanti, M. Montali, A. Mosca, and R. Shraga. Conceptually-grounded mapping patterns for virtual knowledge graphs. *Data Knowledge Engineering*, 145:102157, 2023.
6. J. D’Abramo, A. Zugarini, and P. Torroni. Investigating large language models for text-to-sparql generation. In *Proceedings of the 4th International Workshop on Knowledge-Augmented Methods for Natural Language Processing*, pages 66–80, 2025.
7. M. Lenzerini. Data integration: A theoretical perspective. In *Proceedings of the twenty-first ACM SIGMOD-SIGACT-SIGART symposium on Principles of database systems*, pages 233–246, 2002.
8. S. Migliorini, A. Dalla Vecchia, A. Belussi, and E. Quintarelli. ARTEMIS: a context-aware recommendation system with crowding forecaster for the touristic domain. *Information Systems Frontiers*, pages 1–27, 2024.
9. OpenAIs. Gpt-4 technical report, 2024.
10. C. Pinkel, C. Binnig, E. Jiménez-Ruiz, W. May, D. Ritze, M. G. Skjæveland, A. Solimando, and E. Kharlamov. Rodi: A benchmark for automatic mapping generation in relational-to-ontology data integration. In *European Semantic Web Conference*, pages 21–37. Springer, 2015.
11. A. Sharma, C. Pal, and A. Zouaq. Reducing hallucinations in language model-based SPARQL query generation using post-generation memory retrieval. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4657–4668, 2026.
12. P. Smeros, V. Emonet, R. Wang, A.-C. Sima, and T. M. de Farias. Sparql-llm: Real-time sparql query generation from natural language questions. *arXiv preprint arXiv:2512.14277*, 2025.
13. C. Unger, L. Bühmann, J. Lehmann, A.-C. Ngonga Ngomo, D. Gerber, and P. Cimi-ano. Template-based question answering over RDF data. In *Proceedings of the 21st international conference on World Wide Web*, pages 639–648, 2012.
14. G. Xiao, D. Calvanese, R. Kontchakov, D. Lembo, A. Poggi, R. Rosati, and M. Zakharyashev. Ontology-based data access: A survey. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence IJCAI-18*, pages 5511–5519, 2018.
15. G. Xiao, L. Ren, G. Qi, H. Xue, M. Di Panfilo, and D. Lanti. Llm4vkg: Leveraging large language models for virtual knowledge graph construction. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 4715–4723, 2025.
16. R. Zhao, F. Zhao, and H. Zhang. Correcting on graph: Faithful semantic parsing over knowledge graphs with large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5364–5376, 2025.